Revisiting Pseudo-Rehearsal for Continual Learning with Pretrained Backbones

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Abstract—Anthony Robins's pseudo-rehearsal (Connection Science, 1995) is a continual-learning idea that does not store data and does not train a generator. It feeds random input vectors through the old network, treats the outputs as pseudo-targets, and mixes those pairs into training on the new task. The method dropped out of the literature. Generative replay and model-inversion replay replaced it. I revisit the original recipe in the regime that now dominates continual learning: a frozen ImageNet-pretrained backbone with a small head trained sequentially. The question is whether the backbone's natural-image prior makes the random pseudo-inputs informative again. The answer is no. On Split-CIFAR-100 with a frozen ViT-S/16 and a single 100-way linear head, the faithful Robins recipe finishes at 0.103 final accuracy across three seeds, against 0.102 for plain sequential fine-tuning and 0.529 for class-balanced experience replay. A 2×2 ablation that swaps the input distribution (uniform random pixels vs. STL-10 natural images) and the teacher source (per-task snapshot vs. a joint-trained oracle head) gives the same null result in every cell. Diagnostics show that uniform random pixels through ViT-S collapse to a shrunken feature-space region whose dominant predicted class is exactly $\arg\max_c(W_c \cdot \mu_{\text{random}} + b_c)$ from the joint head, but the failure persists even when the input distribution does not collapse and the teacher is perfect. An α sweep over 100× of the distillation weight (0.5 to 50) confirms the result is not a tuning artifact: all KD-based methods plateau between 0.10 and 0.17. The real takeaway, for this regime, is that output-only regularization is structurally weak in single-linear-head Class-IL with seen-class masking: in these experiments, only class-balanced experience replay closed a meaningful fraction of the joint-vs-sequential gap. Code: https://github.com/Dexin-Huang/pseudo-rehearsal-pretrained.

I. Introduction

A neural network trained sequentially on a stream of tasks destroys what it learned on earlier tasks. McCloskey and Cohen called this catastrophic forgetting [5], and it is still the central problem of continual learning. Approaches split into three families: regularization (constrain the parameters or outputs), rehearsal (keep some old examples around and replay them), and architectural (allocate new capacity per task). I revisit the oldest rehearsal-without-storage idea, Robins's pseudo-rehearsal [6].

The mechanics are simple. After training on task t, freeze a copy of the network as a teacher. While training on task $t+1$, also sample random inputs r, run them through the teacher to get pseudo-targets $q = f_t(r)$, and mix (r, q) pairs into the new-task batches. The student is nudged to keep producing the teacher's outputs on randomly sampled inputs. There is no stored data and no learned generator. The idea predates GANs and VAEs and predates Shin et al.'s deep generative replay [8] by more than two decades.

Pseudo-rehearsal mostly dropped out of the literature. The conventional explanation is that random inputs to a small from-scratch MLP are off-manifold, so the teacher's response on them is uninformative about the old decision boundary. Modern continual learning instead uses learned generators [8, 2], real-exemplar replay, or model-inversion replay [12, 9]. A separate recent line uses frozen ImageNet-pretrained backbones with small prompts or heads on top [10], where the backbone supplies most of the representation.

That last regime is what makes the Robins revisit interesting now. The question is narrow: in the frozen-pretrained-backbone setting, does Robins's original recipe (uniform random pixels through the same backbone, snapshot teacher, KD onto a small student head) recover most of the benefit of methods that require real past data or a learned generator?

The answer here is no. I run a controlled six-method comparison plus three diagnostic extensions on Split-CIFAR-100 with a frozen ViT-S/16 backbone (Section III). The Robins recipe finishes at average accuracy 0.103, within seed noise of plain sequential fine-tuning (0.102) and well below class-balanced experience replay (0.529). A 2×2 ablation over input distribution and teacher quality plus an α sweep over the KD weight leaves the negative result untouched (Section IV). The mechanism check shows that ViT-S maps uniform random pixels to a tight feature-space region whose dominant predicted class is geometrically determined by $\arg\max_c(W_c \cdot \mu_{\text{random}} + b_c)$. The prediction is class 23. The empirical top class is class 23. They match exactly.

The deeper finding is sharper than the conventional dismissal. Even non-collapsing natural inputs and an oracle teacher do not rescue pseudo-rehearsal at any α tested (Section V). In this regime, output-only regularization is weak: the methods that closed the joint-vs-sequential gap in these experiments were the ones that put real per-example old-class supervision into the training batch.

II. Background

A. Pseudo-rehearsal

Robins introduced pseudo-rehearsal as a fix for catastrophic forgetting in small feedforward networks at a time

when the modern continual-learning toolkit did not exist: no GANs, no VAEs, no large pretrained backbones, no exemplar buffers. The procedure is mechanical [6]:

1) At the end of task t, snapshot the network weights as a frozen teacher f_t.
2) Sample a pool of random input vectors $\{r_i\}$ from a simple distribution. The original paper used uniform values per unit.
3) Run the teacher on the pool to get pseudo-targets $q_i = f_t(r_i)$.
4) During task $t{+}1$, mix (r_i, q_i) pairs into each training batch alongside the new-task data, with a distillation loss on the pseudo-pairs in addition to the usual loss on the new data.

The student is nudged to keep producing the teacher's responses on randomly sampled inputs. No data is stored. No generator is trained. The intuition is that if the random inputs cover enough of the input space, distilling the teacher's behavior on them transfers the old decision boundary. The Connection Science paper reported some benefit on toy MLP sequential tasks. The method was dismissed for deep classifiers because random inputs to a network trained from scratch are far off the data manifold and the teacher's outputs on them carry little information about the old-task decision boundary. Atkinson et al. [1] revived it for deep RL but the line otherwise saw little follow-up.

B. Generative and model-inversion replay

The natural successor to Robins replaces random inputs with samples from a model. Shin et al. [8] train a GAN per task as the generator. Later work uses VAEs or normalizing flows. Yin et al. [12] propose DeepInversion, which inverts classifier features to synthesize replay data. Smith et al. [9] extend this to continual learning as Always-Be-Dreaming. These methods are data-free in the same sense Robins is. They also do substantial extra work, either training a generator or running expensive inversion every task, to produce inputs that lie on or near the data manifold.

C. Other regularization

Elastic Weight Consolidation [3] penalizes parameter drift weighted by the previous task's diagonal Fisher information. The online variant [7] accumulates Fisher across tasks. Learning without Forgetting [4] distills the previous teacher onto the current task's real features. LwF is the closest comparator to pseudo-rehearsal. It differs only in the input distribution used for distillation. Experience replay stores a small buffer of past examples; the class-balanced variant keeps a fixed number per seen class.

D. Pretrained-backbone continual learning

Recent work attaches small prompts to a frozen ImageNet-pretrained ViT and trains those per task [10]. The frozen backbone supplies the representation. The per-task adaptation is small. This regime is what makes the Robins revisit interesting now. The backbone might supply the natural-image prior that random pseudo-inputs need.

III. Method

A. Setting

The dataset is Split-CIFAR-100 in 10 tasks of 10 classes each. Class permutations are part of the seed. Each of three seeds defines an independent grouping. Within each task there are 5,000 train and 1,000 test images.

The backbone is vit_small_patch16_224.augreg_in21k_ft_in1k from timm [11], pretrained on ImageNet-21k and fine-tuned on ImageNet-1k. CIFAR-100 images are upsampled to 224×224 with bilinear interpolation and ImageNet-normalized. The backbone is frozen throughout. CIFAR-100 train/test features, a 100,000 random-pixel pool, and a 100,000-sample STL-10 unlabeled pool are precomputed once. Each "training run" then operates on cached 384-dimensional CLS-token features. A full 10-task continual run takes seconds.

The head is a single linear layer Linear(384, 100) with bias, about 38,500 parameters. The linear-head choice is deliberate. It removes head capacity as a confound. Class-IL evaluation uses argmax over the full 100-way head, restricted to seen classes at the current point.

All cross-entropy and KD losses use additive $-\infty$ masks on unseen-class logits. Sequential training is never asked to predict away classes it has not yet seen as targets. Without this masking, sequential fine-tuning would supply implicit negative gradients on future classes and would introduce a logit-suppression artifact separate from the catastrophic forgetting question.

B. Methods compared

All methods optimize the same linear head with AdamW (lr $= 10^{-3}$, batch 256, 10 epochs per task) on cached features.

Sequential. Default seen-class-masked cross-entropy on current-task data only. Lower bound.

Joint. The same head trained once on all 100 classes. Upper bound.

Class-balanced replay. Buffer of 5 examples per seen class, up to 500 total. Each training batch is half current-task and half buffer at constant total batch size.

Online EWC. Diagonal empirical Fisher computed in closed form for a masked linear-CE head:

$$F_W[c,j] = \frac{1}{N}\sum_i (p_i[c] - \mathbb{1}\{c = y_i\})^2 \cdot x_i[j]^2. \qquad (1)$$

Fisher is accumulated across tasks. One running θ^* is kept.

Pseudo-rehearsal (Robins, faithful). At the start of task $t+1$, snapshot the head as teacher h_t. Per step, sample batch_size random images (pixels uniform $[0,1]$, ImageNet-normalized), pass them through the frozen ViT (cached in advance), run the teacher, restrict to

previously-seen classes (seen minus current), and add $\alpha \cdot \mathrm{KD}(q_{\mathrm{old}}, p_{\mathrm{student}})$ where KD is soft-label cross-entropy in teacher-to-student direction. Default $\alpha = 1.0$.

LwF. Same KD machinery as pseudo-rehearsal but the distillation inputs are the current-task real features [4].

C. 2×2 ablation and α sweep

The first sweep (six methods, three seeds, 18 runs) returns a negative result for pseudo-rehearsal. Two natural follow-up hypotheses remain. Each is tested independently.

The first replaces uniform random pixels with STL-10 unlabeled features through the same ViT (pseudo_natural). STL-10 is 100,000 real natural images, out-of-distribution for CIFAR-100. If pseudo-rehearsal needs natural-image structure in the inputs rather than random noise, this should help.

The second replaces the snapshot teacher with the joint-trained head loaded from a checkpoint (pseudo_oracle). This is information leakage. The oracle has seen all 100 classes. It is not a valid method, only a diagnostic. If pseudo-rehearsal fails because the snapshot teacher accumulates errors during CL, an oracle teacher should rescue it.

The third combines both (pseudo_oracle_natural), giving the best-case for pseudo-rehearsal across the two factors. The second sweep adds 3 new pseudo variants × 3 seeds = 9 runs.

A third sweep addresses the tuning critique. It varies the KD weight $\alpha \in \{0.5, 1, 5, 10, 50\}$ on pseudo, pseudo_oracle, and lwf (45 runs). At sufficiently large α the KD term dominates the new-task CE term. The student must mimic the teacher. For pseudo_oracle this should in principle approach joint accuracy. The question is how the curve looks in between.

D. Mechanism check

The diagnostic story for pseudo-rehearsal's failure should not lean on hand-waving about "off-manifold inputs." I make a sharp prediction. The dominant pseudo-target class on random inputs is determined by the geometry of the (random-feature mean, joint head) pair alone:

$$\hat{c} = \arg\max_{c} \big(W_c \cdot \mu_{\mathrm{random}} + b_c \big). \tag{2}$$

If random features collapse tightly enough (small per-dimension standard deviation around the mean), per-sample variation is negligible. The head's response on the mean random feature predicts the dominant argmax. μ_{random} is computed over the 100,000-sample random pool. $\hat{c}$ is compared against the empirical top class.

E. Diagnostic probes

To characterize how random pixels through ViT land in feature space, I report (i) L_2 norm mean and standard deviation of real vs. random features, (ii) nearest-neighbor cosine similarity from a 5K random sample to a 5K real sample, (iii) predictive entropy of the joint head on real vs. random features, and (iv) the top-class histogram of joint-head predictions over the 100,000-sample random pool.

TABLE I
Final accuracy, forgetting (FGT), and backward transfer (BWT) on Split-CIFAR-100 with a frozen ViT-S/16 and a 100-way linear head, three seeds. FGT and −BWT match because the per-task argmax happens at the diagonal.

Method	ACC	FGT	BWT
Joint (UB)	0.788 ± 0.000	0.000	n/a
Replay (5/class)	0.529 ± 0.004	0.450	−0.450
EWC (online)	0.192 ± 0.010	0.825	−0.825
LwF	0.111 ± 0.012	0.919	−0.919
Pseudo-natural	0.106 ± 0.008	0.923	−0.923
Pseudo-oracle	0.105 ± 0.005	0.925	−0.925
Pseudo (Robins)	0.103 ± 0.005	0.927	−0.927
Sequential (LB)	0.102 ± 0.004	0.928	−0.928
Pseudo-oracle-natural	0.099 ± 0.003	0.932	−0.932

TABLE II
2×2 ablation over the two pseudo-rehearsal factors. All four cells overlap within seed noise.

Teacher source	Random pixels	STL-10 natural
CL snapshot	0.103	0.106
Joint (oracle)	0.105	0.099

IV. Experiments

A. Main comparison

Final accuracy on Split-CIFAR-100, mean ± std over three seeds, is shown in Table I. The 2×2 ablation view is in Table II. The running-average accuracy curves are in Fig. 2.

All four pseudo cells overlap within seed noise. Switching the input distribution from random pixels to STL-10 natural images does not help. Replacing the snapshot teacher with the joint-trained oracle does not help. LwF lands in the same neighborhood. EWC offers a modest gain. It constrains the linear head's drift but cannot supply the missing per-class signal. Only class-balanced experience replay separates from the bottom group. At 0.529 it reaches about two-thirds of the joint upper bound while storing 500 features total.

B. α sweep

Sweeping the KD weight α over two orders of magnitude (Fig. 1, Table III) shows the failure is not a tuning artifact. All three KD methods trace nearly parallel monotonic curves. The oracle teacher provides no meaningful advantage over the snapshot teacher at any α tested. At $\alpha \to \infty$ the KD term dominates CE and the student converges to the teacher's behavior on the pseudo-input distribution; whether that recovers joint accuracy on the real CIFAR test set depends on how well that distribution identifies the joint head's old-class decision boundary (Section V).

TABLE III
α sweep on KD methods, three seeds each. All three methods gain less than 0.06 across two orders of magnitude.

	α=0.5	α=1	α=5	α=10	α=50
LwF	0.110	0.111	0.114	0.117	0.164
Pseudo	0.103	0.103	0.107	0.114	0.144
Pseudo-oracle	0.103	0.105	0.111	0.118	0.142

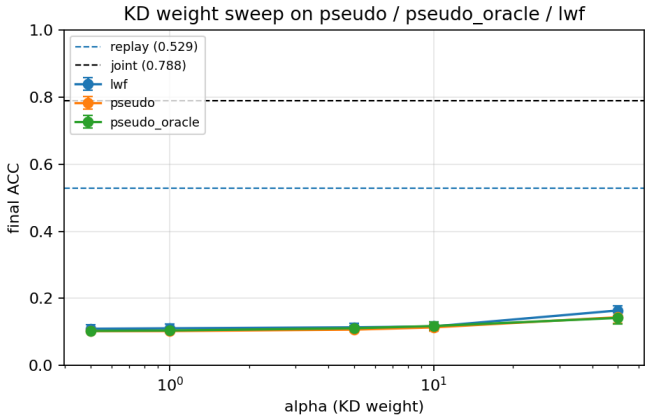

Fig. 1. KD weight sweep on the three distillation methods. Log x-axis. Replay (dashed blue, 0.529) and joint (dashed black, 0.788) are reference lines. All three KD curves crawl from 0.10 to 0.16 across two orders of magnitude. The oracle teacher offers no consistent advantage.

In the explored range the curve has not begun to bend toward joint accuracy. Even at $\alpha = 50$ the gain over the default $\alpha = 1$ is only 0.04 to 0.05.

C. Diagnostics

Diagnostics use the joint head from seed 0 as a fixed teacher on the 100,000-sample random pool. Real-feature L_2 norm is 49.52 $\pm$ 1.70. Random-feature L_2 norm is 34.33 $\pm$ 1.77. Random features sit in a sphere about 31% smaller in radius. Mean nearest-neighbor cosine of random to real is 0.392 (p25 0.381, p75 0.401). Real CIFAR features are far from random-pixel features in direction.

Predictive entropy on real features is 0.513 nats. On random features it is 2.666 nats. Uniform-over-100 reference is 4.605 nats. The teacher is more uncertain on random inputs but not uniformly so. The top-class histogram tells the rest of the story. Class 23 takes 62.7% of all argmax predictions on the random pool. The teacher predicts only 4 of 100 classes anywhere across the 100,000 random inputs. Distribution entropy over predicted classes is 0.664 nats.

D. Mechanism check

The geometric prediction in Eq. 2 holds exactly. $\arg\max_c(W_c \cdot \mu_{\text{random}} + b_c) = 23$. The empirical top class on the random pool is 23. The top-3 predicted classes from the mean feature are $\{23, 60, 76\}$. Class 23 dominates. The per-dimension standard deviation of the random pool, averaged across 384 dimensions, is 0.305. The random-feature distribution is tight enough that per-sample variation does not change the argmax. The random-pixel collapse is fully determined by the joint head and the mean random feature.

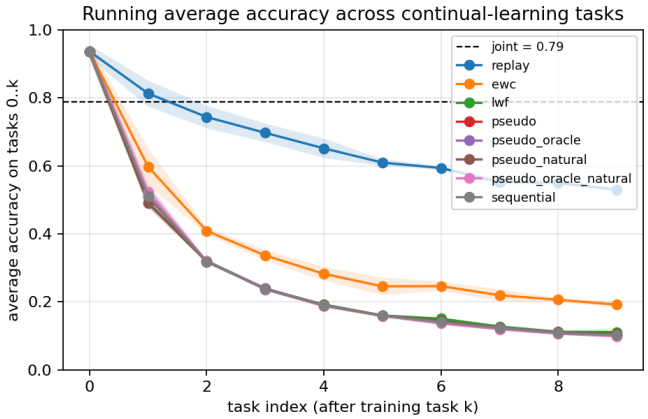

Fig. 2. Running mean accuracy across continual-learning tasks. Replay separates from the pack immediately and decays gracefully. EWC tracks slightly above the bottom group. The four pseudo variants, LwF, and sequential are visually indistinguishable across all 10 tasks.

V. Discussion

A. The mechanism story is sharper than the conventional one

The standard dismissal of Robins is that random inputs are off-manifold and the teacher is uninformative on them. The diagnostics confirm this. Random pixels through ViT-S land in a tight shrunken region of feature space. The teacher classifies 62.7% of them as one dominant class. The mechanism check goes further. The collapse is geometrically predictable from the mean random feature and the head's weights. Robins's recipe at the unit of analysis was always going to give the student degenerate pseudo-soft-targets, a near-constant signal pointing at one arbitrary class.

B. The conventional story is not the whole story

The 2×2 ablation eliminates two natural rescue stories. STL-10 images are real natural images and produce non-degenerate teacher distributions on this backbone. They do not move the result. The oracle teacher, the joint classifier with full knowledge of all 100 classes, also does not move it. Both factors combined still fail. The off-manifold collapse alone is not the full diagnosis, at least under this input choice and this optimization budget. STL-10 is a single out-of-distribution natural-image source, not a guarantee that no natural-image distribution could help; an exhaustive input-distribution sweep is left to future work.

C. A more local diagnosis

In single-linear-head Class-IL with seen-class masking, every training step on task t supplies per-class cross-entropy gradients on 5,000 fresh examples per epoch. The head is aggressively rewritten toward the current task's 10 classes. KD on previously-seen classes adds a counter-pressure on the old-class logits. It acts only through the outputs, not through the per-example supervision the new task gets. Even at $\alpha = 50$ the KD term does not match the CE term's gradient mass without effectively shutting down new-task learning. This generalizes a known observation about LwF's weakness in Class-IL, where the head develops a recency bias. The extension here is that the weakness survives the rescue stories tested: a different input distribution and a perfect teacher.

Replay works for the symmetric reason. It injects real per-example old-class supervision into the batch, on the same footing as new-task examples. EWC partially works because it constrains the head's weights from drifting at all, an approach independent of class-level signal. Output-only regularization in this regime sits between the two: it neither freezes the parameters nor supplies per-example old-class gradient.

D. Connection to data-free continual learning

The finding sharpens what data-free CL needs. DeepInversion [12] and ABD [9] are both data-free in Robins's sense. They also do substantial extra work to produce inputs that lie on the data manifold and elicit class-discriminative responses from the teacher. The implicit assumption is that the input distribution is what was missing from Robins. This paper suggests that, at least in the linear-head Class-IL regime, the input distribution is not the bottleneck. The structural separation between output-only and example-level supervision is. Whether the same holds when the backbone is fine-tunable, or when the head has more capacity, is left to future work.

E. Limitations

The paper studies one backbone (ViT-S/16, ImageNet-21k $\rightarrow$ 1k), one dataset (Split-CIFAR-100), one continual setting (Class-IL, 10 tasks $\times$ 10 classes), one head (linear), one feature-precomputation regime (frozen backbone), three seeds, and one out-of-distribution natural-image source (STL-10). A CLIP or DINO backbone could place random features in a different feature-space region. A multi-layer MLP head or a fine-tunable backbone introduces representational degrees of freedom that EWC and LwF were originally designed to protect. The α sweep stops at 50. At $\alpha \rightarrow \infty$ pseudo_oracle approaches joint accuracy by abandoning new-task learning, but only under the assumption that KD on the chosen pseudo-input distribution identifies the joint head's old-class decision boundary; output matching on off-manifold features need not constrain CIFAR class boundaries even in the limit. The replay number depends on storing 5 examples per class, small but not zero. A fair no-storage floor for pseudo-rehearsal is plain sequential FT, which is what it matches. The paper does not include a class-mean / prototype replay comparator (one stored vector per seen class, no per-example exemplars). Such a comparator would sharpen the distinction the discussion draws between output-only regularization and example-level supervision; it is the most direct experimental follow-up.

VI. Conclusion

The question was whether a frozen ImageNet-pretrained backbone makes Robins's 1995 pseudo-rehearsal competitive again, in the regime where that question is most natural. The answer here is no, across three rescue stories: a single natural-image input source (STL-10), an oracle teacher, and a 100$\times$ sweep of the KD weight. The random-pixel collapse in feature space is real and geometrically predictable, but it is not the only obstacle: STL-10 inputs and an oracle teacher do not move the number either. In this regime, output-only regularization is weak compared to methods that put per-example old-class supervision in the batch. The cheapest competitive method in these experiments stores 5 features per seen class.

Code and Reproducibility

All code, configs, cached features, and result JSON files are at https://github.com/Dexin-Huang/pseudo-rehearsal-pretrained. The reported numbers come from 72 training runs (six baseline methods, three extension methods, α sweep on three methods $\times$ five α values), three seeds each, on RTX 4090 secure-cloud pods. Total compute was under one hour of GPU time. Mean-aggregation and FGT/BWT formulas are in evaluate.py. The mechanism check is in mechanism_check.py. Plots are in plots.py and analyze_alpha.py.

References

[1] Craig Atkinson, Brendan McCane, Lech Szymanski, and Anthony Robins. Pseudo-rehearsal: Achieving deep reinforcement learning without catastrophic forgetting. arXiv preprint arXiv:1812.02464, 2018.

[2] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: a strong, simple baseline. In Advances in Neural Information Processing Systems, 2020.

[3] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. Proceedings of the National Academy of Sciences, 114(13): 3521–3526, 2017.

[4] Zhizhong Li and Derek Hoiem. Learning without forgetting. IEEE Transactions on Pattern Analysis and Machine Intelligence, 40(12):2935–2947, 2017.

[5] Michael McCloskey and Neal J Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. Psychology of Learning and Motivation, 24:109–165, 1989.

[6] Anthony Robins. Catastrophic forgetting, rehearsal and pseudorehearsal. Connection Science, 7(2):123–146, 1995.

[7] Jonathan Schwarz, Wojciech Czarnecki, Jelena Luketina, Agnieszka Grabska-Barwinska, Yee Whye Teh, Razvan Pascanu, and Raia Hadsell. Progress & compress: A scalable framework for continual learning. In International Conference on Machine Learning (ICML), 2018.

[8] Hanul Shin, Jung Kwon Lee, Jaehong Kim, and Jiwon Kim. Continual learning with deep generative replay. In Advances in Neural Information Processing Systems, 2017.

[9] James Smith, Yen-Chang Hsu, Jonathan Balloch, Yilin Shen, Hongxia Jin, and Zsolt Kira. Always be dreaming: A new approach for data-free class-incremental learning. In IEEE/CVF International Conference on Computer Vision (ICCV), 2021.

[10] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. Learning to prompt for continual learning. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

[11] Ross Wightman. PyTorch image models. https://github.com/huggingface/pytorch-image-models, 2019.

[12] Hongxu Yin, Pavlo Molchanov, Jose M Alvarez, Zhizhong Li, Arun Mallya, Derek Hoiem, Niraj K Jha, and Jan Kautz. Dreaming to distill: Data-free knowledge transfer via DeepInversion. In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.